

THE STANDARD INCOMPLETE MARKETS MODEL

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Where we are

- Last lecture: a single household self-insures with a risk-free bond. If $\beta < q$, its assets stay bounded and have a long-run distribution.
- With a continuum of such households, that distribution is the **cross section**, and its mean is the **supply of savings**, $A(r)$.
- Today we close the model: the interest rate adjusts until $A(r)$ meets the demand for savings from outside the household sector. That demand is zero in an exchange economy and equals firms' demand for capital in a production economy.
- Two things are new:
 - the equilibrium has **no closed form**, so we must compute it;
 - the model makes predictions about the **wealth distribution**, which we can confront with data.

1. **Equilibrium:** an exchange economy (*Huggett, 1993*) and a production economy (*Aiyagari, 1994*).
2. **Computation:** a search over the interest rate and, for each rate, two fixed-point problems: the household's policy and the distribution.
3. **Predictions:** what the two models say about interest rates, saving, the wealth distribution and marginal propensities to consume.
4. **An extension:** entrepreneurs, collateral constraints and the top of the wealth distribution.

Part 1: equilibrium

A continuum of households

- A unit mass of households, indexed by $i \in [0, 1]$, each solving the income fluctuation problem.
- Income follows the same Markov chain for everyone, $y \in \{y_1, \dots, y_{N_y}\}$ with transition matrix $\Pi = [\pi_{ij}]$, $\pi_{ij} = \Pr[y' = y_j \mid y = y_i]$, but draws are **independent across households**.
- Let the column vector π^* be the **stationary distribution** of the chain, $\pi^* = \Pi^\top \pi^*$: π_i^* is the long-run probability of y_i .
- All shocks are idiosyncratic: there is **no aggregate risk**.
- A **law of large numbers** applies (Uhlig, 1996): cross-sectional averages equal expectations, so
 - the share of households with income y_i is π_i^* ;
 - aggregates are deterministic, even though every household faces risk.

The household's problem in recursive form

- With incomplete markets, aggregation fails: aggregate saving depends on how wealth is distributed, not only on its total. So the distribution affects **prices**.
- Let λ be the distribution of households over individual states (a, y) . The household's problem is

$$V(a, y; \lambda) = \max_{c, a'} u(c) + \beta \mathbb{E}[V(a', y'; \lambda') \mid y] \quad \text{s.t.} \quad c + q(\lambda) a' = y + a, \quad a' \geq \underline{a}.$$

- For the problem on the right-hand side to be well defined, the household must know everything it needs to solve it:
 - its **individual state**, (a, y) ;
 - today's λ , which sets today's price $q(\lambda)$;
 - tomorrow's λ' , to evaluate the continuation value.
- How does the household know λ' ?

The law of motion of the distribution

- We assume the household knows a law of motion, T , and add it to the problem:

$$\begin{aligned} V(a, y; \lambda) = \max_{c, a'} & u(c) + \beta \mathbb{E}[V(a', y'; \lambda') \mid y] \\ \text{subject to} & c + q(\lambda) a' = y + a, \quad a' \geq \underline{a}, \\ & \lambda' = T(\lambda). \end{aligned}$$

- Without T , the problem is not well defined: the household could not evaluate its continuation value.
- A demanding assumption: the household must forecast how a whole distribution moves.

The distribution is a state variable

- λ is an **aggregate state**: common to all households, and part of every household's state. It is infinite-dimensional, and in principle all of it matters.
- So the well-defined problem is **hard to solve**. The household's state includes a whole distribution, and T must be consistent with the decisions of all households, which themselves depend on T .
- We therefore start with **stationary** equilibria, where $\lambda^* = T(\lambda^*)$. The distribution is still a state, but it does not move, so the household pays attention only to prices.

How the distribution moves: on a grid

- Suppose assets live on a grid $\{a_1, \dots, a_{N_a}\}$ and, for now, that the policy $a^*(a_n, y_i)$ lands on a grid point. A state is a pair (a_n, y_i) .
- The **transition matrix** Q gives the share of households at (a_n, y_i) that move to (a_m, y_j) next period:

$$Q((a_n, y_i), (a_m, y_j)) = \underbrace{\pi_{ij}}_{\text{from } y_i \text{ to } y_j} \underbrace{\mathbf{1}\{a^*(a_n, y_i) = a_m\}}_{\text{does the policy lead to } a_m?}$$

- Next period's distribution adds over today's states:

$$\lambda'(a_m, y_j) = \sum_{n,i} Q((a_n, y_i), (a_m, y_j)) \lambda(a_n, y_i), \quad \text{i.e.} \quad \lambda' = Q^\top \lambda.$$

- So T is a matrix multiplication.

How the distribution moves: in general

- With continuous assets, replace grid points by sets and sums by integrals.
- Q is a **transition function**: the share of households at (a, y_i) that move into a set $\mathcal{A} \times \mathcal{Y}$ next period,

$$Q((a, y_i), \mathcal{A} \times \mathcal{Y}) = \sum_{j: y_j \in \mathcal{Y}} \underbrace{\pi_{ij}}_{\text{from } y_i \text{ to } y_j} \underbrace{\mathbf{1}\{a^*(a, y_i) \in \mathcal{A}\}}_{\text{does the policy lead into } \mathcal{A}?$$

- The law of motion is

$$\lambda'(\mathcal{S}) = T(\lambda)(\mathcal{S}) = \int Q(s, \mathcal{S}) \lambda(ds) \quad \text{for every set } \mathcal{S}.$$

- On a computer, everything is discrete: the previous slide *is* what we compute.

Equilibrium in an exchange economy

- The only asset is a bond issued and held by households: it is in **zero net supply** (Huggett, 1993).
- A household can borrow only if another one lends, so $\underline{a} < 0$ is needed for trade.

Stationary recursive competitive equilibrium

A value function V , policies $c^*(a, y)$ and $a^*(a, y)$, a distribution λ^* and a bond price q such that:

1. given q , V solves the Bellman equation and c^*, a^* are its policies;
2. the bond market clears: $\int a^*(a, y) \lambda^*(da, dy) = 0$;
3. the distribution is stationary: $\lambda^* = T(\lambda^*)$.

Huggett's question

- Huggett (1993) asks: why is the average real risk-free rate so **low**?
 - over 1889–1978 it averaged 0.8% a year (*Mehra and Prescott, 1985*), while calibrated representative-agent economies predict a rate that is too high.
- His conjecture: **market incompleteness** matters. Households cannot insure idiosyncratic risk and can borrow only up to a limit.
- The exchange economy isolates this mechanism. Questions we will answer once we can compute the equilibrium:
 - how far below the rate of time preference does the interest rate fall?
 - how does it depend on the borrowing limit?
 - and on risk aversion?

Adding production

- *Aiyagari (1994)* adds production. Households supply θ efficiency units of labor at wage w , so $y = w \theta$, with θ Markov.
- A representative firm rents capital at rate R and labor at wage w , with $F(K, L)$ constant returns and depreciation δ :

$$R = F_K(K, L), \quad w = F_L(K, L).$$

- A risk-neutral intermediary sells bonds of face value A , collects qA and rents it to the firm. Zero profits require

$$K = q A, \quad (1 + R - \delta) q = 1 \iff R - \delta = r.$$

- Saving now has a use: assets must be **positive** in the aggregate, and $\underline{a} = 0$ is allowed.

Stationary recursive competitive equilibrium

A value function V , policies $c^*(a, \theta)$ and $a^*(a, \theta)$, prices (r, w) and a distribution λ^* such that:

1. given (r, w) , V solves the Bellman equation and c^*, a^* are its policies, with $y = w\theta$ and $q = 1/(1+r)$;
2. aggregates add up individual decisions: $K = q \int a^* d\lambda^*$ and $L = \int \theta d\lambda^*$;
3. prices are marginal products: $r = F_K(K, L) - \delta$ and $w = F_L(K, L)$;
4. the distribution is stationary: $\lambda^* = T(\lambda^*)$.

Aiyagari's question

- Aiyagari (1994) asks: how important is uninsured individual risk for **aggregate saving**?
- It was an open question where capital comes from: life-cycle saving, bequests, or precaution. Several authors conjectured that precautionary saving could account for a large share of the US capital stock (*Zeldes, 1989*).
- Why capital matters for this question: in an exchange economy, saving cannot change aggregate resources. With capital, precautionary saving builds capital and, through diminishing returns, lowers the interest rate.
- Questions we will answer once we can compute the equilibrium:
 - how much higher are capital and the saving rate than with complete markets?
 - how does the answer depend on the size and persistence of earnings risk, and on risk aversion?
 - how unequal is wealth, compared with earnings?

Equilibrium is one equation in one unknown

- Fix r , and let λ_r be the stationary distribution implied by the household's policy at that r . Then r pins down:
 - the **supply of savings**, $A(r) = \int a \, d\lambda_r$;
 - the wage $w(r)$ and the **demand for capital** $K(r)$, from the firm's condition.
- Equilibrium is a zero of excess supply:

$$\text{Huggett: } A(r) = 0, \quad \text{Aiyagari: } \frac{1}{1+r} A(r) = K(r).$$

- Existence: $A(r)$ is continuous, sits near $\underline{a}$ for low r , and diverges as $r \uparrow \beta^{-1} - 1$ (last lecture). In Aiyagari, $K(r)$ decreases in r . The curves cross.
- **Uniqueness is not guaranteed**, though in practice we find a single crossing.

Part 2: computing a stationary equilibrium

How to compute a stationary equilibrium

Search over r

update r until the asset market clears

1. Household problem

given r : a fixed point in value or policies

gives $a^*(a, y)$ on the grid



2. Distribution

given a^* : a fixed point $\lambda = Q^\top \lambda$

gives λ_r and $A(r)$

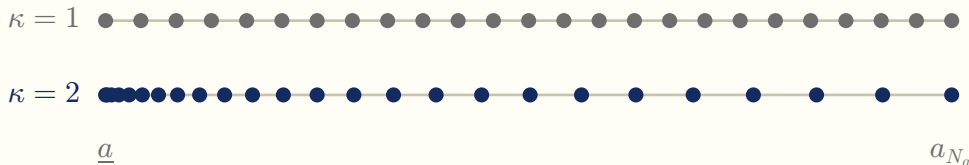
Steps 1 and 2 are fixed-point problems, solved once for every guess of r .

Preliminaries: the asset grid

- A computer cannot handle a continuum of asset levels: we must **discretize** the domain of assets, $[\underline{a}, \infty)$, into a finite grid $\{a_1, \dots, a_{N_a}\}$.
- A numerical solution is then a set of **matrices**: $a^*(a_n, y_i)$, $c^*(a_n, y_i)$ and $\lambda(a_n, y_i)$, of size $N_a \times N_y$.
- The grid starts at $a_1 = \underline{a}$ and ends at an **upper bound** a_{N_a} we must choose:
 - if $\beta < q$, assets are bounded, but we do not know the bound;
 - we pick a value and **check it** after solving (below).

Preliminaries: spacing the grid

- A convenient family: $a_n = \underline{a} + (a_{N_a} - \underline{a}) x_n^\kappa$, with x_n evenly spaced on $[0, 1]$.
- With $\kappa = 1$ the grid is evenly spaced; with $\kappa > 1$ it puts more points near the borrowing limit.



- Ideally, we want more grid points where the functions we are solving for have more **curvature**.

Preliminaries: discretizing income

- Income processes estimated from data have a **continuous support**. We assume an AR(1) in logs as our leading example:

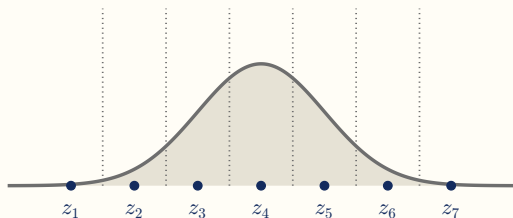
$$z' = \rho z + \varepsilon', \quad \varepsilon' \sim N(0, \sigma_\varepsilon^2), \quad y = \exp(z).$$

- We approximate it with a **Markov chain**, which means choosing two things:
 - a grid of points $\{z_1, \dots, z_N\}$;
 - a transition matrix $\Pi = [\pi_{ij}]$.
- We want the chain to mimic the **stochastic properties** of the process.
- Two standard methods: Tauchen's and Rouwenhorst's (*Tauchen, 1986; Rouwenhorst, 1995*).

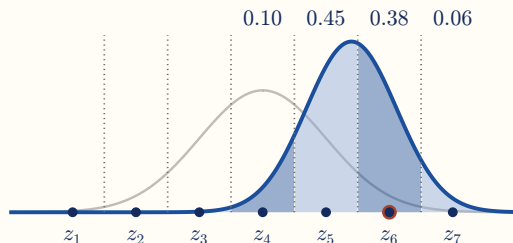
Preliminaries: Tauchen's method

- **Grid:** N evenly spaced points on $\pm m$ unconditional standard deviations, $\sigma_z = \sigma_\varepsilon / \sqrt{1 - \rho^2}$. Each point owns the cell around it.

Long-run distribution of z



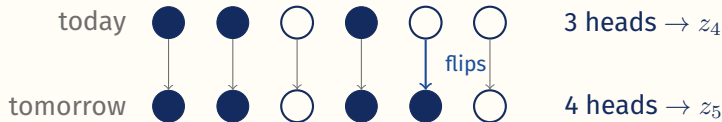
Distribution of z' given z_6



- **Transition matrix:** row i is the conditional density of z' given z_i , $N(\rho z_i, \sigma_\varepsilon^2)$, integrated over each cell. The tails go to z_1 and z_N .

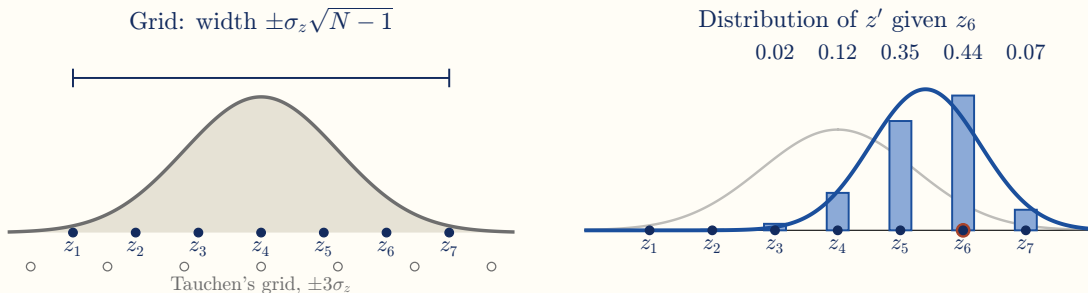
Preliminaries: Rouwenhorst's method, the idea

- Take the grid as given for a moment: N evenly spaced points.
- Take $N - 1$ independent coins. Each period, each coin keeps its face with probability p and flips with probability $1 - p$. The state is the number of heads, $k \in \{0, \dots, N - 1\}$: N values, one for each grid point z_{k+1} .



- Given N , we choose two things to match the AR(1) exactly: p for the **persistence**, and the width of the grid for the **variance**.
 - each coin, and so their sum, has autocorrelation $2p - 1$: set $p = (1 + \rho)/2$;
 - in the long run each coin is heads half the time, so the number of heads is $\text{Binomial}(N - 1, \frac{1}{2})$, and a grid on $\pm \sigma_z \sqrt{N - 1}$ gives variance σ_z^2 (Kopecky and Suen, 2010).

Preliminaries: Rouwenhorst's method, grid and transition matrix



- Left: the grid, evenly spaced on $\pm\sigma_z\sqrt{N-1}$. For $N = 7$ it is narrower than Tauchen's $\pm 3\sigma_z$.
- Right: row 6 of Π . From k heads, tomorrow's heads are the heads that stay, $\text{Binomial}(k, p)$, plus the tails that flip, $\text{Binomial}(N-1-k, 1-p)$. The curve is the AR(1)'s conditional density, scaled to the grid step.
- Instead of the normal areas that Tauchen's method uses, the chain matches the **persistence** and the **variance** exactly.

Output: which method to use

Persistence and standard deviation implied by each chain (target: $\sigma_z = 0.3$)

True ρ	N	Tauchen ($m = 3$)		Rouwenhorst	
		ρ	σ_z	ρ	σ_z
0.50	5	0.499	0.335	0.500	0.300
0.50	9	0.499	0.308	0.500	0.300
0.90	5	0.932	0.381	0.900	0.300
0.90	9	0.898	0.331	0.900	0.300
0.98	5	1.000	0.404	0.980	0.300
0.98	9	0.989	0.374	0.980	0.300

- Tauchen overstates risk, by more as persistence rises. For $\rho \geq 0.9$, the range that earnings estimates deliver, use **Rouwenhorst**.

A test economy for the household problem

- To compare algorithms we fix the price and solve one household problem:
 - CRRA utility with $\sigma = 2$, $\beta = 0.96$, $r = 3\%$ (so $\beta < q$), and $\underline{a} = 0$;
 - log income AR(1) with $\rho = 0.9$ and $\sigma_z = 0.3$, Rouwenhorst with $N_y = 7$.
- We measure accuracy with **Euler equation errors** (Judd, 1992): off the grid, at points where the limit is slack,

$$EE(a, y) = \log_{10} \left| 1 - \frac{u'^{-1}(\frac{\beta}{q} \mathbb{E}[u'(c^*(a', y')) \mid y])}{c^*(a, y)} \right|.$$

- $EE = -3$ means a mistake of \$1 for every \$1,000 spent.

Method 1: value function iteration

Algorithm: value function iteration on the grid

1. Guess $V_0(a_n, y_i)$.
2. For every (a_n, y_i) , choose a' on the grid:

$$V_{k+1}(a_n, y_i) = \max_{a_m \geq \underline{a}} u(y_i + a_n - q a_m) + \beta \sum_j \pi_{ij} V_k(a_m, y_j).$$

3. Stop when $\|V_{k+1} - V_k\| < \varepsilon$; otherwise, go back to step 2.

- Robust: it needs neither concavity nor differentiability, and it always converges, since the Bellman operator is a contraction.
- But it is **slow and coarse**: the error shrinks by a factor β per iteration; each iteration evaluates N_a choices at each of $N_a N_y$ states, about $N_a^2 N_y$ operations; and the policy is a step function.

Speeding up VFI: Howard's improvement

- The expensive step is the maximization. But the **policy** converges long before the value function does.
- Howard's improvement: after each maximization, keep the policy $a^*(a_n, y_i)$ fixed and update V H times,

$$V(a_n, y_i) \leftarrow u(y_i + a_n - q a^*(a_n, y_i)) + \beta \sum_j \pi_{ij} V(a^*(a_n, y_i), y_j),$$

which needs no maximization.

- With $H = 50$, the number of maximizations falls from 443 to 15, for the same answer.
- But Howard does not improve accuracy: the policy still lives on the grid.

Value function iteration with a continuous choice

- Let a' take any value in $[\underline{a}, (y_i + a_n)/q)$, not only grid points. Two new ingredients:
 - **interpolate** the continuation value $\mathbb{E}[V_k(a', y') \mid y_i]$ between grid points;
 - **maximize** at every state with a one-dimensional optimizer, e.g. golden-section search.
- What it buys: the policy is no longer a staircase, so accuracy is no longer tied to the grid spacing.
- What it costs: $N_a N_y$ maximizations per iteration, each with dozens of evaluations of an interpolated function. Howard steps help here too.
- **Golden-section search assumes a single peak**: it needs a concave problem.

Method 2: policy function iteration

- Iterate on the Euler equation, not the Bellman equation.

Algorithm: policy function iteration

1. Guess a consumption policy $c_0(a_n, y_i)$.
2. For every (a_n, y_i) , find the a' , any value, that solves

$$u'(y_i + a_n - q a') = \frac{\beta}{q} \sum_j \pi_{ij} u'(c_k(a', y_j)),$$

interpolating c_k between grid points. If the left side exceeds the right at $a' = \underline{a}$, set $a' = \underline{a}$.

3. Set $c_{k+1} = y_i + a_n - q a'$ and iterate until $\|c_{k+1} - c_k\| < \varepsilon$.

- Accurate, but each iteration solves $N_a N_y$ nonlinear equations: a root finder inside a loop.

Method 3: the endogenous grid method

- The idea of Carroll (2006): fix **tomorrow's** assets on the grid and solve for **today's**.
- If $a' = a_m$ is a grid point, the right side of the Euler equation needs no interpolation, and the Euler equation **can be inverted in closed form**:

$$\tilde{c}(a_m, y_i) = u'^{-1} \left(\frac{\beta}{q} \sum_j \pi_{ij} u'(c_k(a_m, y_j)) \right).$$

- The budget constraint then tells us **which assets today lead to that choice**:

$$\tilde{a}(a_m, y_i) = \tilde{c}(a_m, y_i) + q a_m - y_i.$$

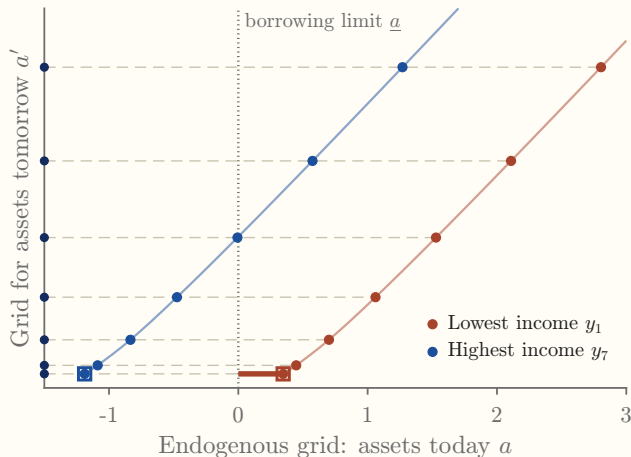
- The points $\tilde{a}$ form an **endogenous grid**: no root finding and no maximization, only function evaluations.

The endogenous grid method: the algorithm

Algorithm: endogenous grid method

1. Guess $c_0(a_n, y_i)$, e.g. $c_0 = y_i + a_n - q \underline{a}$.
 2. For every (a_m, y_i) , compute $\tilde{c}(a_m, y_i)$ and $\tilde{a}(a_m, y_i)$.
 3. For every y_i , interpolate the pairs $(\tilde{a}(a_m, y_i), a_m)$ to get $a'(a_n, y_i)$ on the original grid.
 4. Where $a_n < \tilde{a}(a_1, y_i)$, below the lowest point of the endogenous grid, the limit binds: set $a'(a_n, y_i) = \underline{a}$.
 5. Set $c_{k+1} = y_i + a_n - q a'$ and iterate until $\|c_{k+1} - c_k\| < \varepsilon$.
- Why step 4: at $\tilde{a}(a_1, y_i)$ the Euler equation holds exactly with $a' = \underline{a}$. With less wealth, the household would like to borrow more, but cannot.

Output: the endogenous grid



- Each grid point for a' , on the vertical axis, maps into a point $\tilde{a}$ of the endogenous grid.
- The square marks $\tilde{a}(a_1, y_i)$. Between the borrowing limit and the square, the limit binds and $a' = \underline{a}$.
- With low income the limit binds up to $a = 0.35$; with high income it never binds.

Output: the race

Method	N_a	Iterations	Time (s)	Euler error, $\log_{10}$	
				Mean	95th pct.
VFI	1,000	443	3.59	-2.4	-2.0
VFI + Howard	1,000	15	0.27	-2.4	-2.0
VFI, continuous	200	443	3.28	-6.1	-5.1
VFI, continuous + Howard	200	17	0.20	-6.1	-5.1
Policy function iteration	200	382	1.52	-6.9	-5.2
EGM	200	382	0.12	-6.9	-5.2
EGM	1,000	381	0.18	-8.3	-6.5

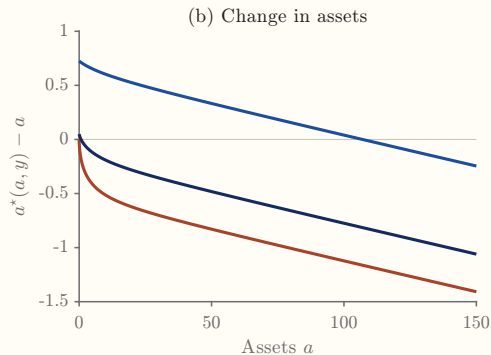
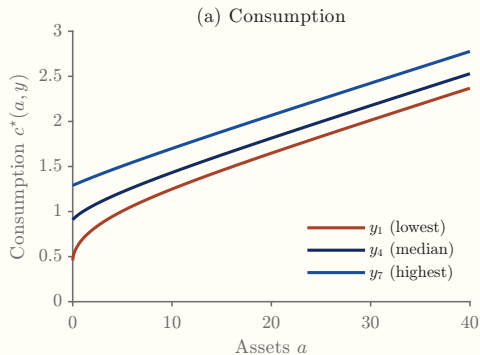
- With a continuous choice, VFI is accurate too. On the same grid, policy function iteration is faster: finding a zero is cheaper than maximizing.
- EGM gives the same answer as policy function iteration, 13 times faster. It is as fast as grid-based VFI with Howard, and **about a million times more accurate**.

MATLAB on a laptop; tolerance 10^{-8} ; errors on 5,000 points off the grid.

When EGM is not enough

- EGM wins when the problem has one endogenous state, one continuous choice, a concave value function and an invertible marginal utility.
- Adding exogenous states, such as more income points, is cheap: the cost grows only linearly.
- It struggles with:
 - more than one endogenous state, e.g. two assets: the endogenous grid comes out as scattered points;
 - more than one continuous choice, e.g. consumption and hours;
 - non-concave problems, e.g. discrete choices such as default or occupation (*Iskhakov et al., 2017*);
 - no closed-form inverse of marginal utility: we are back to a root finder.
- Then we fall back on value function iteration, on the grid if the problem is not concave, or on policy function iteration, which is still accurate.

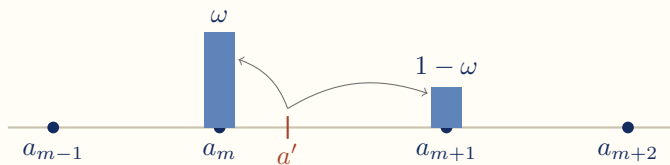
Output: policies, and a check on the grid



- Assets fall above some level for **every** income state, so the upper bound is not binding.
- The highest-income household keeps saving up to $a \approx 107$: an upper bound of $a_{N_a} = 40$ would have **distorted its policy**.

The distribution: two ways to compute it

- **Monte Carlo**: simulate many households for many periods with the policy, and use the cross section at the end.
 - simple, but noisy: the error falls only with the square root of the number of households.
- **Histogram**: track the mass at every grid point. A household whose choice a' falls between two grid points is split between them by a lottery:



$$\omega = \frac{a_{m+1} - a'}{a_{m+1} - a_m}$$

- The closer grid point gets more mass, and the mean is exact:
 $\omega a_m + (1 - \omega) a_{m+1} = a'$. The histogram has no sampling noise.

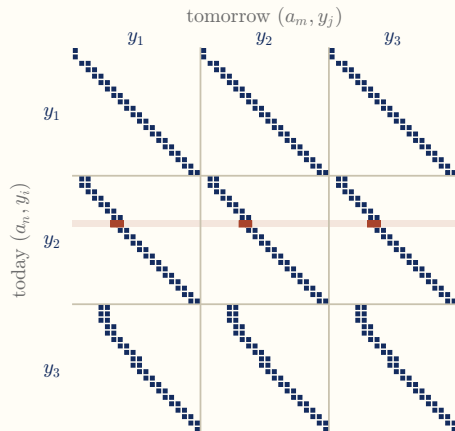
The histogram method: the transition matrix

- Order the states as $s = (a_n, y_i)$ and fill an $N_a N_y \times N_a N_y$ **sparse** matrix:

$$Q((a_n, y_i), (a_m, y_j)) = \pi_{ij} \omega_{nim},$$

where ω_{nim} is the lottery weight that $a^*(a_n, y_i)$ puts on a_m .

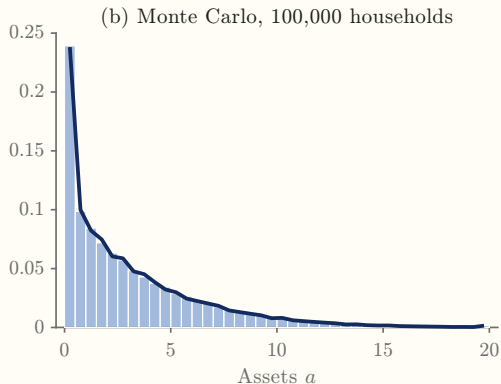
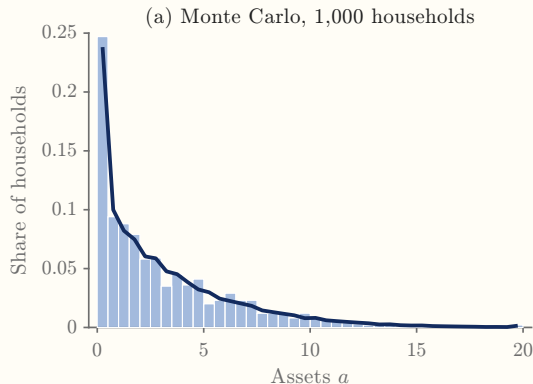
- Each row has at most $2N_y$ nonzero entries, as in the highlighted row: two neighboring grid points for each income state tomorrow.



The histogram method: the stationary distribution

- The stationary distribution solves $\lambda = Q^\top \lambda$. Three ways to find it:
 - iterate $\lambda_{k+1} = Q^\top \lambda_k$ from any guess (827 iterations here);
 - compute the eigenvector of $Q^\top$ for the unit eigenvalue;
 - solve the linear system $(Q^\top - I)\lambda = 0$, with one equation replaced by $\sum_s \lambda_s = 1$.
- All three take a fraction of a second.

Output: histogram against Monte Carlo



- Line: histogram method, mean assets 3.114, in 0.1 seconds. With 1,000 households, Monte Carlo gives 3.072; with 100,000, 3.109 in 3 seconds.
- Noise in $A(r)$ is deadly in the search over r : it makes excess supply jagged in r .

The search over r : clearing the market

- Excess supply $A(r)$ (Huggett) or $qA(r) - K(r)$ (Aiyagari) is one function of one variable.
- **Bisection** on r is slow but safe. The bracket:
 - from above, $r < \beta^{-1} - 1$, where $A(r)$ diverges;
 - from below, $r > -\delta$ in Aiyagari. Under an ad hoc limit, the household problem is well defined for $r \leq 0$ too, so in Huggett the bracket can include negative rates.
- Alternatives in Aiyagari: iterate on K , updating $K_{k+1} = (1 - \xi)K_k + \xi qA(r(K_k))$ with damping ξ ; or a secant method.
- **Warm starts**: start each household solution from the previous c . Late in the bisection the price barely moves, so the household problem converges in a few iterations.

The whole algorithm

Algorithm: stationary equilibrium

0. Build the asset grid and the income chain (y, Π) . Set a bracket $[r_L, r_H]$.
 1. Set $r = (r_L + r_H)/2$, $q = 1/(1 + r)$, and $w(r)$ from the firm's condition.
 2. Solve the household problem by EGM, starting from the last c : get $a^*(a_n, y_i)$.
 3. Build Q with lottery weights and solve $\lambda = Q^\top \lambda$.
 4. Compute excess supply, $E(r) = q \sum_{n,i} a^*(a_n, y_i) \lambda(a_n, y_i) - K(r)$.
 5. If $|E(r)| < \varepsilon$, stop. If $E(r) > 0$, set $r_H = r$; otherwise set $r_L = r$. Go back to step 1.
- For Huggett, set $K(r) = 0$ and $w = 1$.

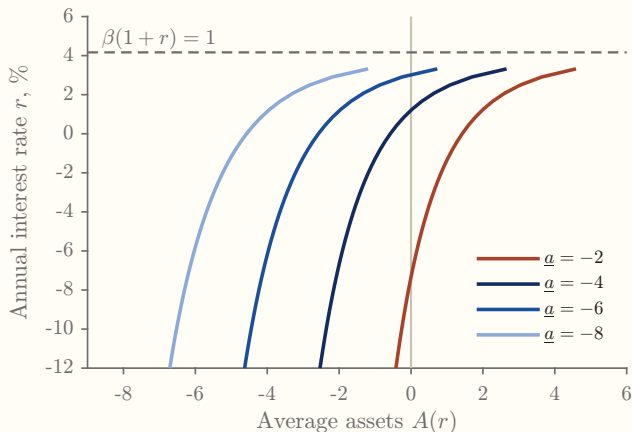
A checklist

- **Tolerances:** tighter for the household problem and the distribution than for the search over r . Otherwise, their numerical noise flips the sign of excess supply.
- **Top of the grid:** $a^*(a, y) < a$ near a_{N_a} for every y , and no mass at a_{N_a} .
- **Accuracy:** Euler errors off the grid.
- **Market clearing:** report excess supply at the solution, and check Walras' law: the goods market clears too.
- **Replicate** a known result before trusting a new one. We do this next.

Part 3: Huggett (1993) and the risk-free rate

- Back to Huggett's question: how much does self-insurance depress the risk-free rate?
- A model period is two months. Income is an employment state:
 - $y \in \{0.1, 1\}$: not employed and employed;
 - $\pi(1 | 1) = 0.925$ and $\pi(1 | 0.1) = 0.5$: a spell without work lasts two periods on average;
 - the standard deviation of annual earnings is 20% of the mean.
- $\beta = 0.99322$ (0.96 per year), CRRA with $\sigma = 1.5$, and limits $\underline{a} \in \{-2, -4, -6, -8\}$. Annual income is 5.3, so $\underline{a} = -5.3$ is one year of income.
- With complete markets, the annual rate is $\beta^{-6} - 1 = 4.2\%$.

Prediction: the supply of savings



- Each curve is $A(r)$ for one limit; equilibrium is where it crosses $A = 0$.
- A looser limit shifts $A(r)$ **left**: households borrow more at every rate, so the rate must rise to clear the market.
- All curves bend up toward $\beta(1+r) = 1$.

$\sigma = 1.5$. Rates are annual. With an ad hoc limit, $A(r)$ is well defined for $r \leq 0$, i.e. $q \geq 1$.

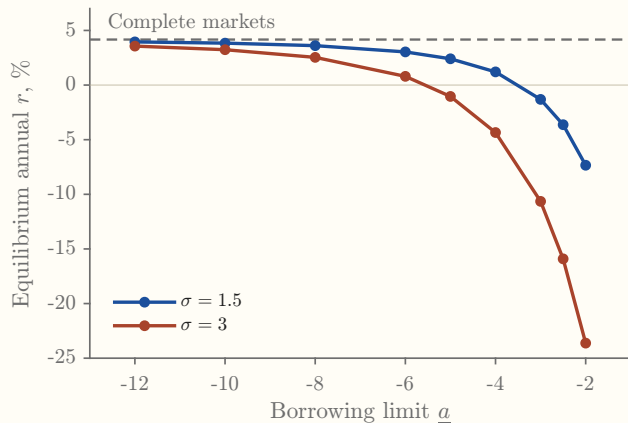
Replication: Huggett's Table 1

Equilibrium annual interest rate, %				
Borrowing limit $\underline{a}$	-2	-4	-6	-8
$\sigma = 1.5$				
Huggett (1993)	-7.1	2.3	3.4	4.0
This lecture	-7.3	1.2	3.0	3.6
$\sigma = 3$				
Huggett (1993)	-23.1	-2.7	1.8	3.7
This lecture	-23.6	-4.3	0.8	2.5

- Same ranking, same orders of magnitude. Our rates are lower, and the gap **survives a finer grid**: at Huggett's price for $\underline{a} = -4$, our excess supply is 0.87, far above his tolerance of 0.0025.
- A useful habit: when a replication misses, find out whether the gap comes from your code or from theirs.

Huggett's rates for $\sigma = 3$ are computed from his reported prices.

Prediction: tighter limits, lower rates



- As the limit loosens, the rate approaches the complete-markets rate, but stays **below** it.
- As it tightens, the rate falls fast: the less households can borrow, the lower the rate must be to stop them from saving.
- More risk aversion means more precaution, and lower rates.

All limits shown are ad hoc: the natural limit, $-y_{\min}/(1 - q)$, lies below -15 at every rate shown.

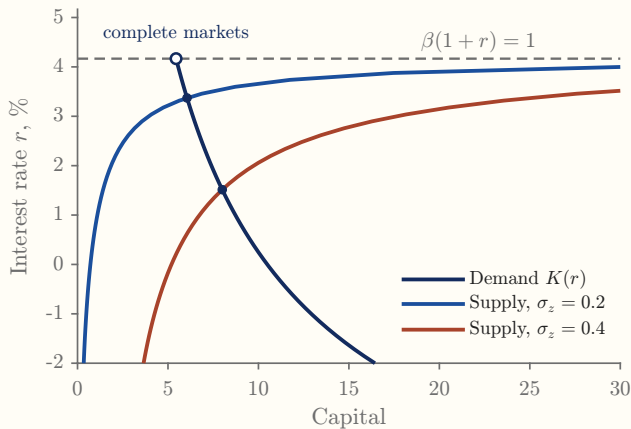
What we learn from Huggett

- Uninsurable risk lowers the risk-free rate below the rate of time preference, $\beta(1 + r^*) < 1$.
- The size of the effect depends on how tightly households can borrow: with one year of income as a limit, the rate falls by about 1.5 points.
- A low risk-free rate can come from **market incompleteness**, without implausible preferences.
- Next: what changes when saving can build capital.

Part 4: Aiyagari (1994), saving and capital

- Back to Aiyagari's question: how much does self-insurance raise aggregate saving?
- Annual model. Cobb–Douglas technology, $F(K, L) = K^\alpha L^{1-\alpha}$, with $\alpha = 0.36$ and $\delta = 0.08$; $\beta = 0.96$; $\underline{a} = 0$.
- Log labor efficiency is an AR(1) with persistence $\rho \in \{0, 0.3, 0.6, 0.9\}$ and standard deviation $\sigma_z \in \{0.2, 0.4\}$. Relative risk aversion $\sigma \in \{1, 3, 5\}$.
- Aiyagari uses Tauchen with seven states; so do we, to replicate him.
- With complete markets, $r = \beta^{-1} - 1 = 4.17\%$ and the saving rate is $s = \delta\alpha/(r + \delta) = 23.7\%$.

Prediction: savings and capital



Tauchen, $\rho = 0.9$, $\sigma = 3$, as in Aiyagari.

- Demand $K(r)$ decreases with r ; supply $qA(r)$ increases and diverges as $\beta(1+r) \rightarrow 1$.
- With complete markets, supply would be flat at $\beta(1+r) = 1$: the open dot, $K = 5.45$.
- With risk, capital is **higher** and the rate **lower**:
 $K = 6.05$ and $r = 3.37\%$ ($\sigma_z = 0.2$); $K = 8.00$ and $r = 1.51\%$ ($\sigma_z = 0.4$).

Replication: Aiyagari's Table II

Equilibrium interest rate, % ($\sigma_z = 0.4$)						
ρ	This lecture			Aiyagari (1994)		
	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
0.0	4.06	3.78	3.45	4.06	3.78	3.42
0.3	3.98	3.49	2.93	3.96	3.42	2.80
0.6	3.80	2.91	2.00	3.76	2.78	1.81
0.9	3.40	1.51	-0.09	3.31	1.29	-0.35

- Close everywhere, and **almost identical** when shocks are not persistent; the gaps grow with ρ and σ .
- Relative to 23.7% with complete markets, incomplete markets raise the saving rate by 0.2 points ($\rho = 0, \sigma = 1$) to 12.7 points ($\rho = 0.9, \sigma = 5$). Aiyagari: 0.2 to 14.0.

The panel with $\sigma_z = 0.2$ matches within 0.15 points.

Output: the discretization matters

Equilibrium interest rate, %

	$\sigma_z = 0.2$			$\sigma_z = 0.4$		
	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$	$\sigma = 1$	$\sigma = 3$	$\sigma = 5$
$\rho = 0.9$						
Tauchen, 7 states	3.95	3.37	2.67	3.40	1.51	-0.09
Rouwenhorst, 7 states	4.01	3.58	3.05	3.59	2.12	0.73

- Tauchen's chain is riskier than the AR(1) it approximates (first table of Part 2), so it **overstates** precautionary saving.
- With persistent shocks and high risk aversion, the bias reaches **0.8 percentage points** of interest rate: as large as the effects we want to measure.
- From now on: Rouwenhorst, $\rho = 0.9$, $\sigma_z = 0.4$, $\sigma = 3$. Then $r = 2.12\%$ and $K/Y = 3.56$.
- The answers depend heavily on ρ . We discuss how persistent income really is when we study inequality.

How much more do households save?

- In a steady state, investment replaces depreciation, and the firm sets $F_K = r + \delta$. With Cobb–Douglas,

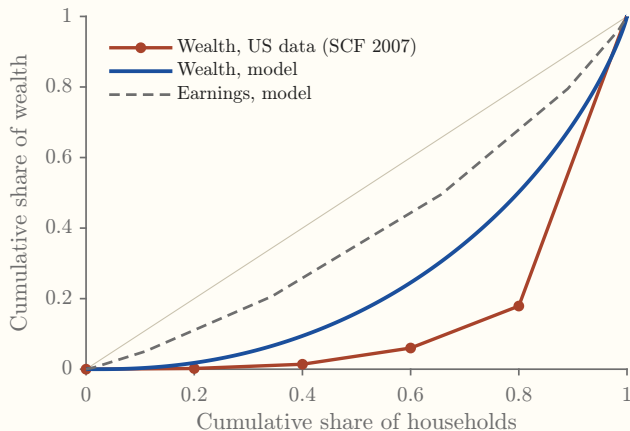
$$s = \frac{\delta K}{Y} = \frac{\delta \alpha}{r + \delta}.$$

- The formula is the same with complete and incomplete markets. Risk changes saving **only through** r :

$$s^{IM} - s^{CM} = \frac{\delta \alpha (\beta^{-1} - (1 + r^{IM}))}{(r^{IM} + \delta)(\beta^{-1} - 1 + \delta)} > 0.$$

- For moderate risk, the increase is at most about one point. It is large only with very persistent shocks and high risk aversion.
- The model's lasting influence comes from its other predictions: the **distribution of wealth**, which Aiyagari himself notes is less unequal than in the data, and, in later work, **marginal propensities to consume**. We turn to these next.

Prediction: the distribution of wealth



Data: Survey of Consumer Finances 2007, from Krueger, Mitman and Perri (2016).

- The model turns an earnings Gini of 0.22 into a wealth Gini of **0.48**: precautionary saving amplifies inequality.
- But the data show a wealth Gini of 0.78.
- The gap is at the **top**: the richest 1% hold 4.7% of wealth in the model, and 33.5% in the data.

Prediction: the wealth distribution in numbers

	Share of wealth held by quintile, %					Top 1%	Gini
	Q1	Q2	Q3	Q4	Q5		
US data (SCF 2007)	0.2	1.2	4.6	11.9	82.5	33.5	0.78
Aiyagari economy	1.8	7.6	15.1	25.7	49.8	4.7	0.48

- The bottom 40% hold almost nothing in the data; in the model, they hold 9%.
- The middle class is too rich, and the rich are **not rich enough**.
- Why? Households save to self-insure. Once they hold a buffer, the motive fades, and with $\beta(1+r) < 1$ they stop accumulating. Nothing in the model makes the rich keep saving.

Prediction: marginal propensities to consume



- A **windfall** is a one-time, unexpected transfer Δ added to today's assets. The annual MPC out of it is $[c^*(a + \Delta, y) - c^*(a, y)]/\Delta \approx \partial c^*/\partial a$, with Δ equal to 1% of the wage.
- Households at the limit consume all of a windfall; the MPC then falls quickly toward the permanent income value.
- Few households are near the limit, so the mean annual MPC is only 0.08, well below the 0.15–0.25 per *quarter* found for tax rebates (*Kaplan and Violante, 2014*).

What we learn from Aiyagari

- Self-insurance raises capital and lowers the interest rate; the effect on the saving rate is modest unless risk is large and persistent.
- The model generates a wealth distribution from first principles, and it is more unequal than earnings.
- Two of its predictions are far from the data:
 - less wealth at the top;
 - fewer households with little wealth, hence a lower average MPC.
- The second matters for fiscal and monetary policy, later in the course. Next, we extend the model and look again at the first.

Part 5: an extension with entrepreneurs

Why entrepreneurs

- In US data, families that own and run a business are about 14% of families. They earn 22% of income and hold 40% of wealth (*Quadrini, 2000*).
- They are concentrated at the **top** of the wealth distribution.
- Running a firm gives rich households a reason to keep saving: if borrowing is limited, a large firm needs the owner's own wealth, and the return on it is high.
- We add entrepreneurs to the Aiyagari economy, keep everything else the same, and solve for the new equilibrium.

Entrepreneurs: the environment

- Each household has, besides θ , an entrepreneurial ability z , drawn from a Pareto distribution; each year, with some probability, it draws a new one.
- At the start of each period, the household chooses to work, for $w \theta$, or to run a firm, for profits

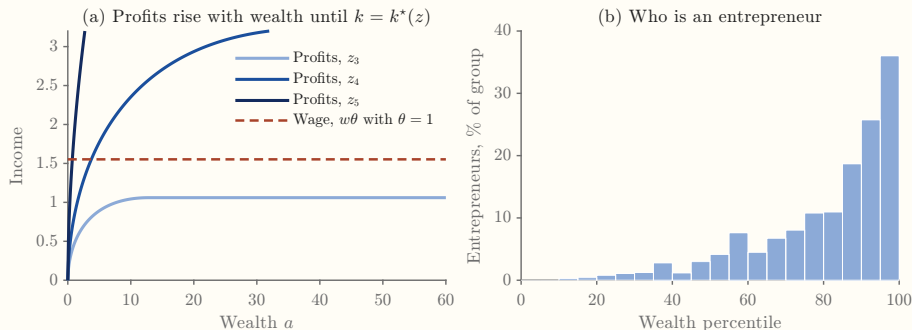
$$\pi(a, z) = \max_{k \leq \phi a, l} z (k^\alpha l^{1-\alpha})^{1-\nu} - w l - (r + \delta) k.$$

- Two features: **decreasing returns** cap the size of each firm; a **collateral constraint**, $k \leq \phi a$, ties it to the owner's wealth.
- The budget constraint becomes $c + q a' = a + \max\{w \theta, \pi(a, z)\}$.
- A corporate sector, with the Aiyagari technology, hires the remaining capital and labor, so $w = w(r)$ as before.

Entrepreneurs: what changes in the algorithm

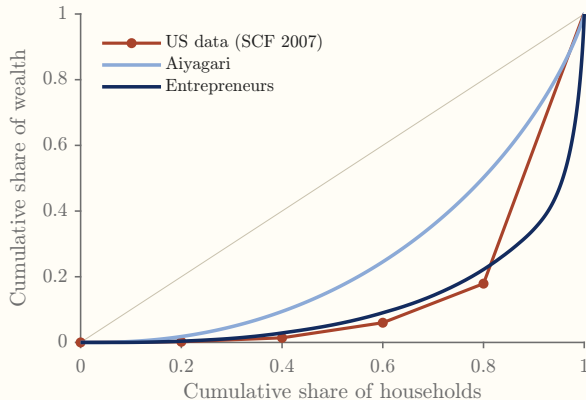
- Income $\max\{w\theta, \pi(a, z)\}$ has a **convex kink** where the household switches to entrepreneurship, so V need not be concave in a .
- Then the Euler equation is necessary but **not sufficient**: EGM may return a point that satisfies it but is not optimal, as warned in Part 2.
- Options:
 - VFI with Howard steps: it does not need concavity. We use it here, with $N_a = 300$ and 35 income-ability states.
 - An extension of EGM that compares the candidate points (*Iskhakov et al., 2017*).
- The search over r is unchanged: bisection. The capital market now clears with $q \int a^* d\lambda = \int k d\lambda + K_c$.
- A lesson: the right algorithm depends on the model, and robust methods are worth keeping in the toolbox.

Prediction: who becomes an entrepreneur



- Profits rise with wealth until the firm reaches its unconstrained size. Able but poor households must work, or run small firms, until they **save their way out** of the constraint.
- In a back-of-the-envelope calibration, entrepreneurs are 7% of households, hold 29% of wealth, run 35% of capital, and are 36% of the richest 5% (data: 14% of families, 40% of wealth).

Prediction: wealth inequality



- Entrepreneurs bend the Lorenz curve toward the data, mostly by emptying the middle.
- In this calibration, the wealth Gini is 0.48 without entrepreneurs, 0.75 with them and 0.78 in the data; the top 1% share is 4.7%, 22.9% and 33.5%.
- The interest rate falls, from 2.12% to -0.55% .
- What happens as collateral constraints loosen, and why the interest rate falls: Problem Set 2.

Summary

- In a stationary equilibrium with idiosyncratic risk, the distribution is constant, and equilibrium reduces to one price that clears one market.
- To compute it, search over r and, for each r , solve the household problem and find the distribution. Use Rouwenhorst for persistent income, EGM for the household, the histogram method for the distribution, and bisection for the price.
- Self-insurance lowers the interest rate (Huggett) and raises capital (Aiyagari), but barely moves the saving rate unless earnings risk is large and persistent.
- The benchmark predicts wealth more unequal than earnings, but less concentrated at the **top** than in the data, and too few households near the limit, hence a low average MPC.
- With entrepreneurs who face collateral constraints, the model predicts a top tail close to the data.
- Next lecture: transitional dynamics and aggregate risk.

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